1 / QUESTION, DATA AND TRANSFER

Can a small adaptation help?

Question. Can a small trained head on frozen TorNet radar features improve current tornado detection? We compare the current scan with five-minute history, stronger warning-negative weighting and a nonlinear head. Warning researchers benefit from knowing which changes improve discrimination and which increase missed cases.

Contribution. We reuse a public detector's learned features, train new prediction heads and run controlled comparisons. This is an adaptation study, not a new weather foundation model. It classifies tornado presence at the current scan; it does not predict future tornadoes.

Public data and defensible holdouts

MIT TorNet v1.1 contains 203,133 catalog clips in TOR, WRN and NUL sampling categories. We used 2014-2015 authors' source-test cases. Removing shared source-training episodes/events and nearby source-training records left 6,414 planned clips. Actual NetCDF QA retained 6,414; 0 exclusions applied to every configuration. The target is the supplied binary terminal-frame label, never the sampling category.

Actual-label split	Clips	Positive	Groups	Positive groups
train	3667	438	28	12
val	1325	177	10	8
test	1422	89	10	8

Complete episodes sharing any input-window UTC date stay together. The seed-42 metadata-only split was fixed before loading labels or scores. Actual time, label, shape, episode/event/date isolation and source-archive MD5 checks passed. Training fits all scalers and coefficients; validation selects epochs, configuration and thresholds; final testing happens after a hashed decision lock.

A genuine pre-fit QA failure exposed 2,325 WRN files with legacy embedded event/episode IDs. The authors document corrected warning IDs in v1.1. Filenames match the corrected catalog IDs, and actual terminal timestamps match the catalog exactly. We use catalog IDs for grouping, retain embedded IDs as diagnostics, and keep strict ID equality for TOR/NUL. Frame labels and the predeclared split are unchanged; the failed audit is preserved.

What was pretrained, and what did we train?

The public TorNet CNN has 4,665,409 parameters. We froze every backbone weight and disabled dropout, then spatially max-pooled dropout_11 into 384 features per frame. Inputs retain six physical radar fields, two tilts, range-fold mask and the original range/inverse-range coordinates. Normalization and NaN filling are already inside the checkpoint. Our inputs and original logits matched the authors' loader within the verified tolerances; repeated features were identical.

Model revision b4d103aa...; upstream code 92d9d97b...; complete hashes and versions are in the provenance receipt. Data archives declare CC BY 4.0; TorNet software is MIT licensed. We retain MIT Lincoln Laboratory attribution and its release notice. The checkpoint lacks a sample list. 1,667/6,414 planned clips share a terminal date with source-training records; date overlap does not establish duplicate samples, but entirely unseen weather is not guaranteed.

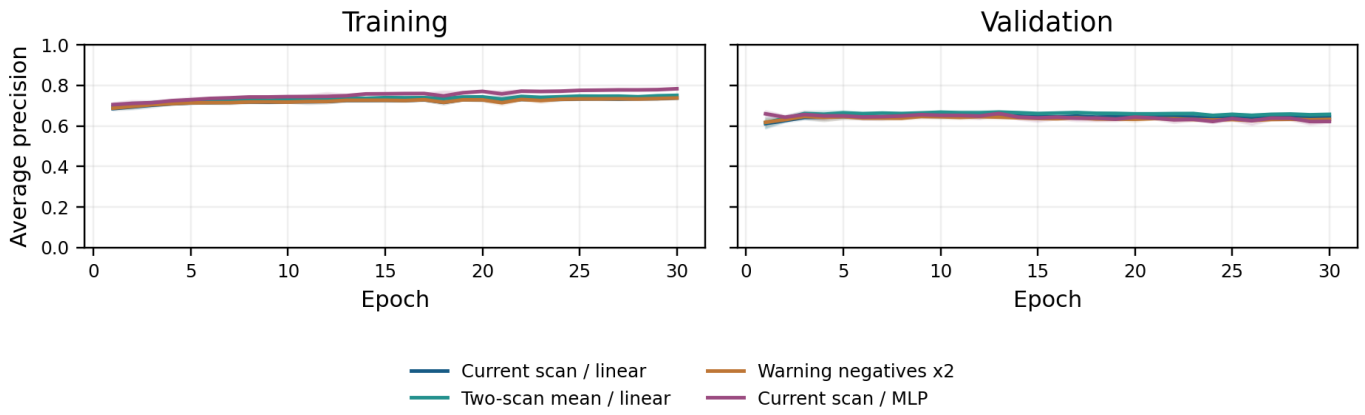
2 / CONTROLLED EXPERIMENTS

One change, three repeated seeds

Each experiment starts from the trained current-scan linear baseline. History averages current and previous standardized features; hard negatives double only WRN label-zero loss weights; the MLP adds 64 ReLU units and dropout 0.2. No experimental changes are combined.

Configuration	Head params	Mean val AP ± SD	Ensemble AP
Current scan / linear	385	0.658 ± 0.005	0.660
Two-scan mean / linear	385	0.675 ± 0.001	0.676
Warning negatives x2	385	0.656 ± 0.005	0.656
Current scan / MLP	24,705	0.670 ± 0.007	0.674

Seeds 42/43/44; AdamW lr 0.001, weight decay 0.01; batch 64; 30 epochs. Train-only class-balanced BCE weights are normalized to mean one. All linear heads use the same initialization and batch order within each seed. Keep the earliest maximum-validation-AP epoch. AP summarizes precision across recall thresholds and is more informative than accuracy with uncommon positives.



Training and validation AP: mean over three seeds, shading ±1 standard deviation. The best seed is not selected. Weighted training BCE and unweighted validation BCE are separately plotted in the notebook and figure files.

A real less-successful attempt

Warning negatives x2 was a genuine less-successful adaptation. Giving non-tornadic warning examples more weight might reduce false alarms. Mean validation AP was 0.656, versus 0.658 for the baseline (difference -0.003). Warning labels are uncertain; extra negative pressure can also suppress real tornado cues. The authors instead downweighted warning cases. This is a plausible explanation, not an established mechanism. We kept the predeclared comparison and selected the final configuration using all three seeds, without adding the unsuccessful change to other runs. At each model's validation-recall threshold, warning-negative FPR was 0.530 versus 0.507 for baseline; this secondary measure can trade off with overall ranking quality. The AP difference is small relative to seed variation. Rechoosing a threshold to preserve 80% recall can offset a loss-weight change's effect on raw scores, so extra negative weight does not guarantee fewer false alarms.

Frozen-feature extraction took 201.0 s; the longest head fit took 1.5 s. The two full archives required 32.46 GB of transfer, a substantial data cost despite inexpensive head training. Timings are observed on this machine, not guarantees for other hardware.

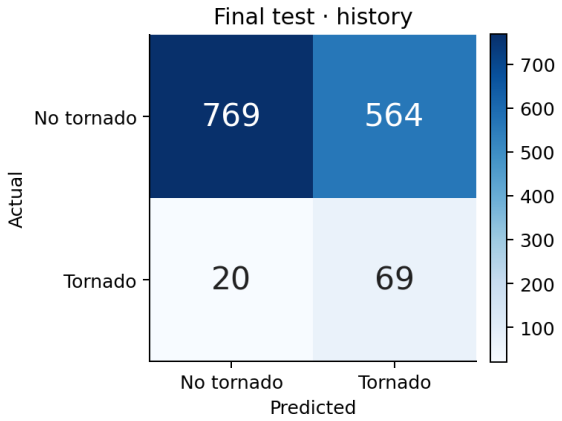
3 / DECISION AND FINAL EVIDENCE

A holdout used after selection

Selected adaptation: Two-scan mean / linear. We ranked mean validation AP across three seeds. Candidates within 0.01 AP of the best favor fewer parameters, then one frame, then declared order. The selected model averages three saved heads' sigmoid scores. Its locked threshold is 0.2967. Each scored reference gets its own highest observed validation threshold achieving at least 80% recall, with ties retained.

Final-test model	AP	Precision	Recall	F1	WRN FPR
Two-scan mean / linear	0.548	0.109	0.775	0.191	0.550
Training majority	0.063	0.000	0.000	0.000	0.000
Unchanged TorNet CNN	0.517	0.198	0.764	0.314	0.387
Radar-statistics logistic	0.142	0.086	0.876	0.156	0.858
Current scan / linear	0.508	0.109	0.787	0.192	0.545

Validation prevalence 0.134; test prevalence 0.063; majority accuracy 0.937; selected accuracy 0.589. AP depends on prevalence, so compare models within each split. The physical-statistics logistic reference uses eight fixed radar summaries and train-only imputation/scaling. The unchanged CNN reference retains its original spatial prediction head. Scores are not calibrated operational probabilities.



Selected-model confusion counts: TN 769, FP 564, FN 20, TP 69. Validation sensitivity targets need not carry over to test storms. WRN FPR uses only warning-category cases whose stored target is zero.

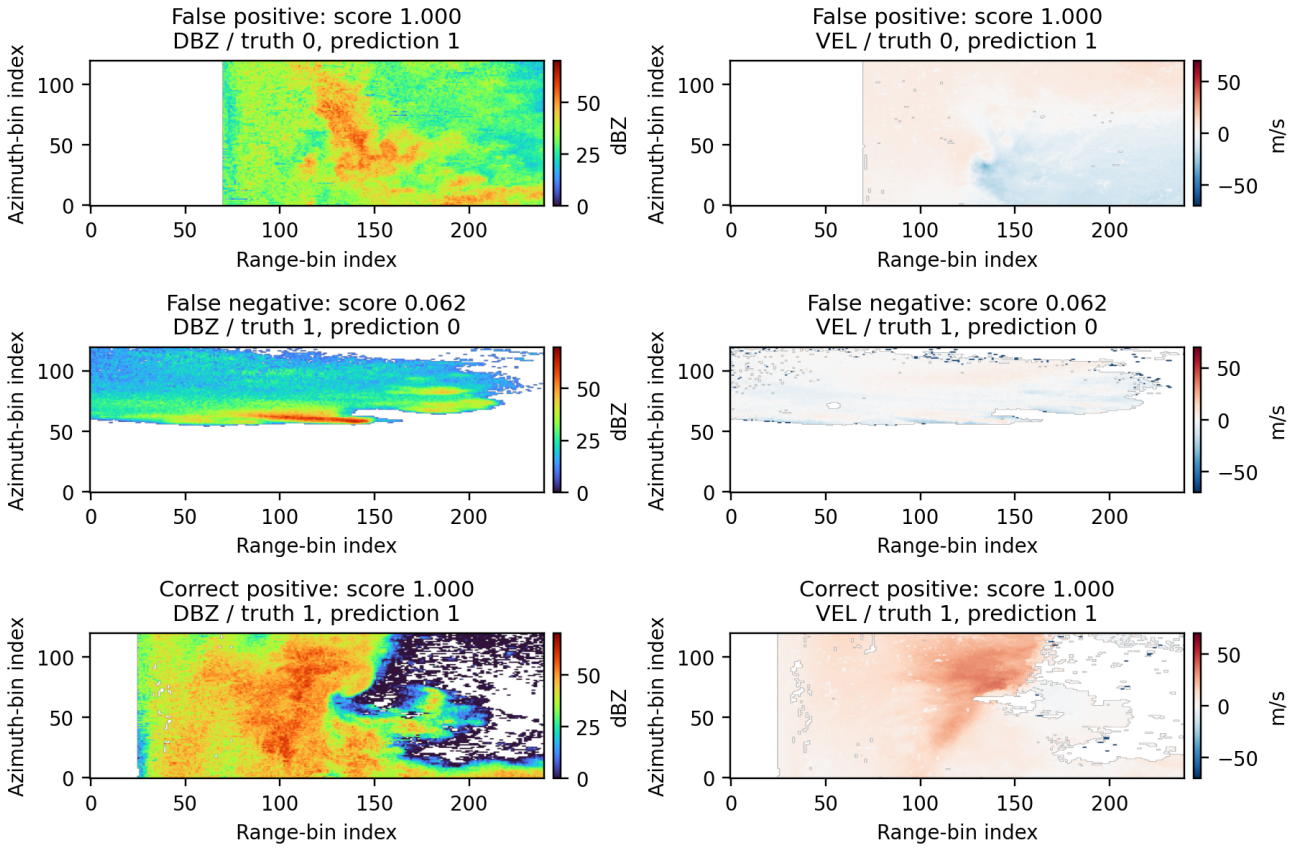
Selected AP's descriptive 95% storm-group bootstrap interval: [0.360, 0.658], 1000/1000 valid paired draws (seed 42). Only 10 test groups are available, and one planned group contains 703 clips. Per-group metrics and paired differences accompany the notebook; individual clips are not independent trials.

Paired test AP difference, adaptation minus unchanged CNN: +0.031; 95% group-bootstrap interval [-0.061, 0.155]. This comparison uses the same resampled storm groups for both models.

The selected adaptation exceeded the unchanged CNN's test AP in this particular grouped holdout. The paired interval includes zero. At the locked validation-derived thresholds, the adaptation detected 69 positive clips versus 68 for the unchanged CNN, but produced 564 false positives versus 276. AP describes overall ranking, while these counts describe one operating point. We retain the validation-selected configuration as this experiment's final adaptation; the evidence does not support warning deployment.

4 / ERRORS, LIMITS AND REPRODUCTION

What the evidence can support



Lowest tilt in native polar array coordinates. Examples are the highest-score false positive, lowest-score false negative and a seeded correct positive when present. DBZ shows reflectivity; VEL shows radial velocity. These are illustrations of saved predictions, not proof of why a decision was wrong.

Limits. Retrospective event-centered chips differ from searching live radar. This is current detection, not advance forecasting. Only two years and few independent storms were studied; curated prevalence differs from real warning operations. Warning targets can be uncertain. Source-training exclusion follows the authors' protocol, while other episodes on the same weather days may have been seen during pretraining. Temporal averaging tests smoothing, not directional storm evolution.

Reproduce. run_study.py orchestrates archive verification, audit, feature extraction, twelve fits, decision locking and final evaluation. project.ipynb was executed with visible results. The pre-run JSON remains unchanged; completion is documented by the run receipts. Identical reruns validate and reuse sealed artifacts; fresh experiments require a separate study copy. Python 3.11.9; torch 2.8.0+cu128; Keras 3.11.3; NumPy 2.2.6; scikit-learn 1.7.2; hardware NVIDIA GeForce RTX 5060 Ti. Full settings, hashes, versions and run logs are included.

References and attribution

A Benchmark Dataset for Tornado Detection and Prediction using Full-Resolution Polarimetric Weather Radar Data (2025)

Authors' accessible preprint

TorNet code and distribution notice

Public pretrained checkpoint

2014 v1.1 data / CC BY 4.0

2015 v1.1 data / CC BY 4.0

Technical draft for team review. Both members must understand the complete project; personal statements, signatures and private peer evaluations must reflect their own participation. No teamwork evidence is inferred from the automated experiment.